

Lesson 2 · Homework

NCEA Level 2 · AS 91262 · Turning points and optimisation

Time: about 60 minutes

Answer version · tutor's working, not an official mark scheme

Show all working. Justify every maximum or minimum. [A] Achieved, [M] Merit, [E] Excellence.

1. Find the turning point of $y = x^2 + 6x - 7$ and state whether it is a maximum or a minimum. [A]

$$\frac{dy}{dx} = 2x + 6 = 0, \text{ so } x = -3; y = 9 - 18 - 7 = -16$$

$$\frac{d^2y}{dx^2} = 2 > 0, \text{ so } (-3, -16) \text{ is a minimum}$$

2. Find the turning point of $y = 3 + 10x - x^2$ and state its nature. [A]

$$\frac{dy}{dx} = 10 - 2x = 0, \text{ so } x = 5; y = 3 + 50 - 25 = 28$$

$$\frac{d^2y}{dx^2} = -2 < 0, \text{ so } (5, 28) \text{ is a maximum}$$

3. Find the turning points of $f(x) = x^3 - 3x + 2$ and justify their nature. [A]

$$f'(x) = 3x^2 - 3 = 0, \text{ so } x = 1 \text{ or } x = -1$$

$$f(1) = 1 - 3 + 2 = 0; f(-1) = -1 + 3 + 2 = 4$$

$$f''(x) = 6x: f''(1) = 6 > 0, \text{ so } (1, 0) \text{ is a minimum}$$

$$f''(-1) = -6 < 0, \text{ so } (-1, 4) \text{ is a maximum}$$

4. Find the turning points of $f(x) = 2x^3 + 3x^2 - 36x + 1$ and justify their nature. [A]

$$f'(x) = 6x^2 + 6x - 36 = 6(x + 3)(x - 2) = 0, \text{ so } x = -3 \text{ or } x = 2$$

$$f(-3) = -54 + 27 + 108 + 1 = 82; f(2) = 16 + 12 - 72 + 1 = -43$$

$$f''(x) = 12x + 6: f''(-3) = -30 < 0, \text{ so } (-3, 82) \text{ is a maximum}$$

$$f''(2) = 30 > 0, \text{ so } (2, -43) \text{ is a minimum}$$

5. Find the three turning points of $y = x^4 - 8x^2$ and state their nature. [A]

$$\frac{dy}{dx} = 4x^3 - 16x = 4x(x^2 - 4) = 0, \text{ so } x = 0, x = 2 \text{ or } x = -2$$

$$y(0) = 0; y(\pm 2) = 16 - 32 = -16$$

$$\frac{d^2y}{dx^2} = 12x^2 - 16: \text{ at } x = 0 \text{ it is } -16 < 0, \text{ so } (0, 0) \text{ is a maximum}$$

at $x = \pm 2$ it is $32 > 0$, so $(2, -16)$ and $(-2, -16)$ are minimums

6. For which values of x is $f(x) = x^2 - 4x + 9$ decreasing? [A]

$$f'(x) = 2x - 4 < 0, \text{ so } x < 2$$

7. For which values of x is $f(x) = x^3 - 3x^2 + 5$ increasing? [A]

$$f'(x) = 3x^2 - 6x = 3x(x - 2), \text{ zero at } x = 0 \text{ and } x = 2$$

Up-parabola, positive outside its roots: $x < 0$ or $x > 2$

8. For which values of x is $f(x) = \frac{1}{3}x^3 - x^2 - 8x$ decreasing? [A]

$$f'(x) = x^2 - 2x - 8 = (x - 4)(x + 2), \text{ zero at } x = 4 \text{ and } x = -2$$

Negative between the roots: $-2 < x < 4$

9. The curve $y = x^2 + bx + 5$ has a turning point at $x = 3$. Find b and the y -coordinate of the turning point. [M]

$$\frac{dy}{dx} = 2x + b = 0 \text{ at } x = 3: 6 + b = 0, \text{ so } b = -6$$

$$y = 9 - 18 + 5 = -4; \text{ the turning point is } (3, -4)$$

10. A model rocket's height is $h = 40t - 5t^2$ metres after t seconds. Find its maximum height. [A]

$$\frac{dh}{dt} = 40 - 10t = 0, \text{ so } t = 4 \text{ s}$$

$$h = 160 - 80 = 80; h'' = -10 < 0, \text{ so maximum height } 80 \text{ m}$$

11. The daily cost of making x units is $C = 2x^2 - 80x + 1000$ dollars. Find the number of units that gives the minimum cost, and the minimum cost. [A]

$$\frac{dC}{dx} = 4x - 80 = 0, \text{ so } x = 20 \text{ units}$$

$$C = 800 - 1600 + 1000 = 200; C'' = 4 > 0, \text{ so the minimum cost is } \$200$$

12. Two positive numbers add to 20. Use calculus to find the largest possible value of their product. [M]

Numbers x and $20 - x$: $P = x(20 - x) = 20x - x^2$

$P' = 20 - 2x = 0$, so $x = 10$; $P'' = -2 < 0$, so maximum

Largest product $= 10 \times 10 = 100$

13. A rectangular garden against a house wall is fenced on the other three sides with 40 m of fencing. Find the dimensions that give the largest area, and that area. [M]

Sides x , x and $40 - 2x$: $A = x(40 - 2x) = 40x - 2x^2$

$A' = 40 - 4x = 0$, so $x = 10$; $A'' = -4 < 0$, so maximum

Garden 10 m by 20 m; area 200 m^2

14. **Extension** A rectangle is drawn under the curve $y = 12 - x^2$ with two corners on the x -axis and two corners on the curve, symmetrical about the y -axis. Find the largest possible area of the rectangle, and justify that it is a maximum. [E]

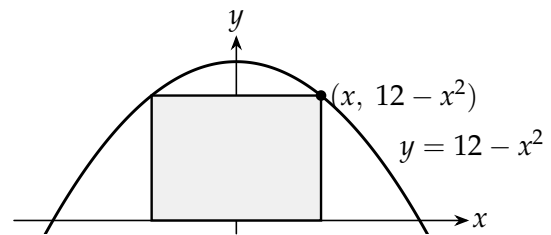
Corner $(x, 12 - x^2)$: width $2x$, height $12 - x^2$

$A = 2x(12 - x^2) = 24x - 2x^3$

$A' = 24 - 6x^2 = 0$, so $x^2 = 4$, $x = 2$ ($x > 0$)

$A'' = -12x = -24 < 0$, so maximum

Largest area $= 2(2)(12 - 4) = 32$ square units



Optional extra: StudyTime Level 2 Calculus Guide, "Calculating Turning Points" from page 16.